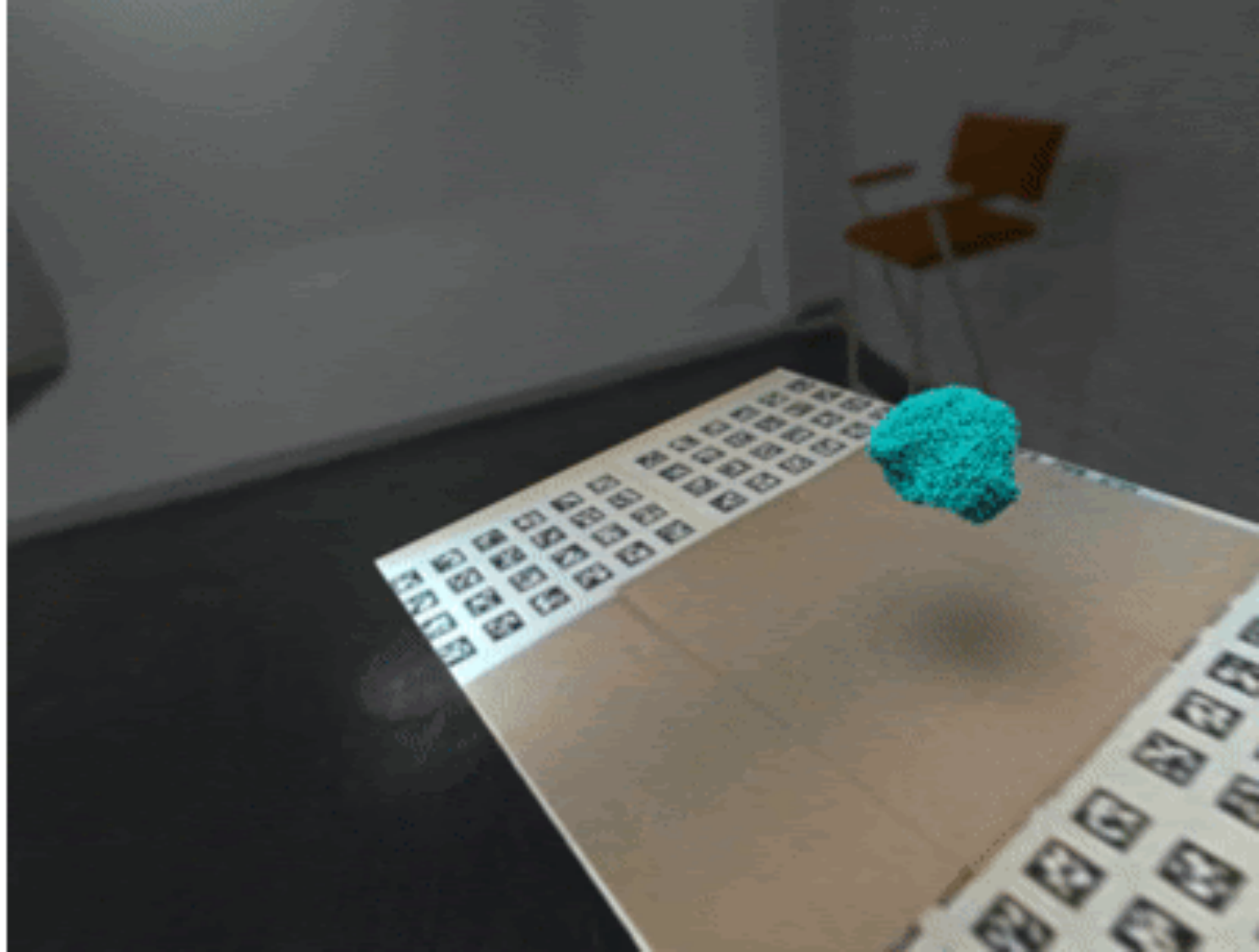
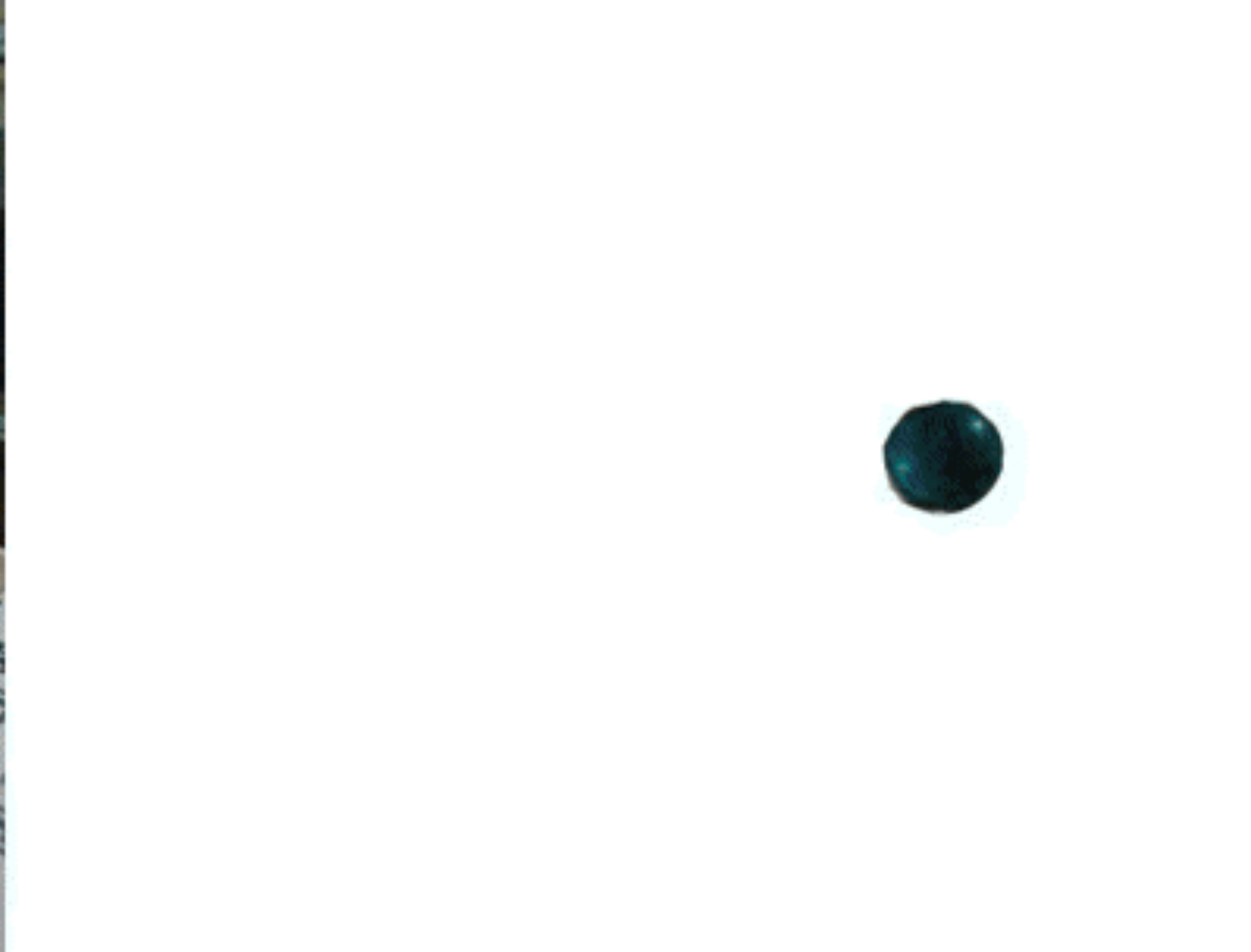
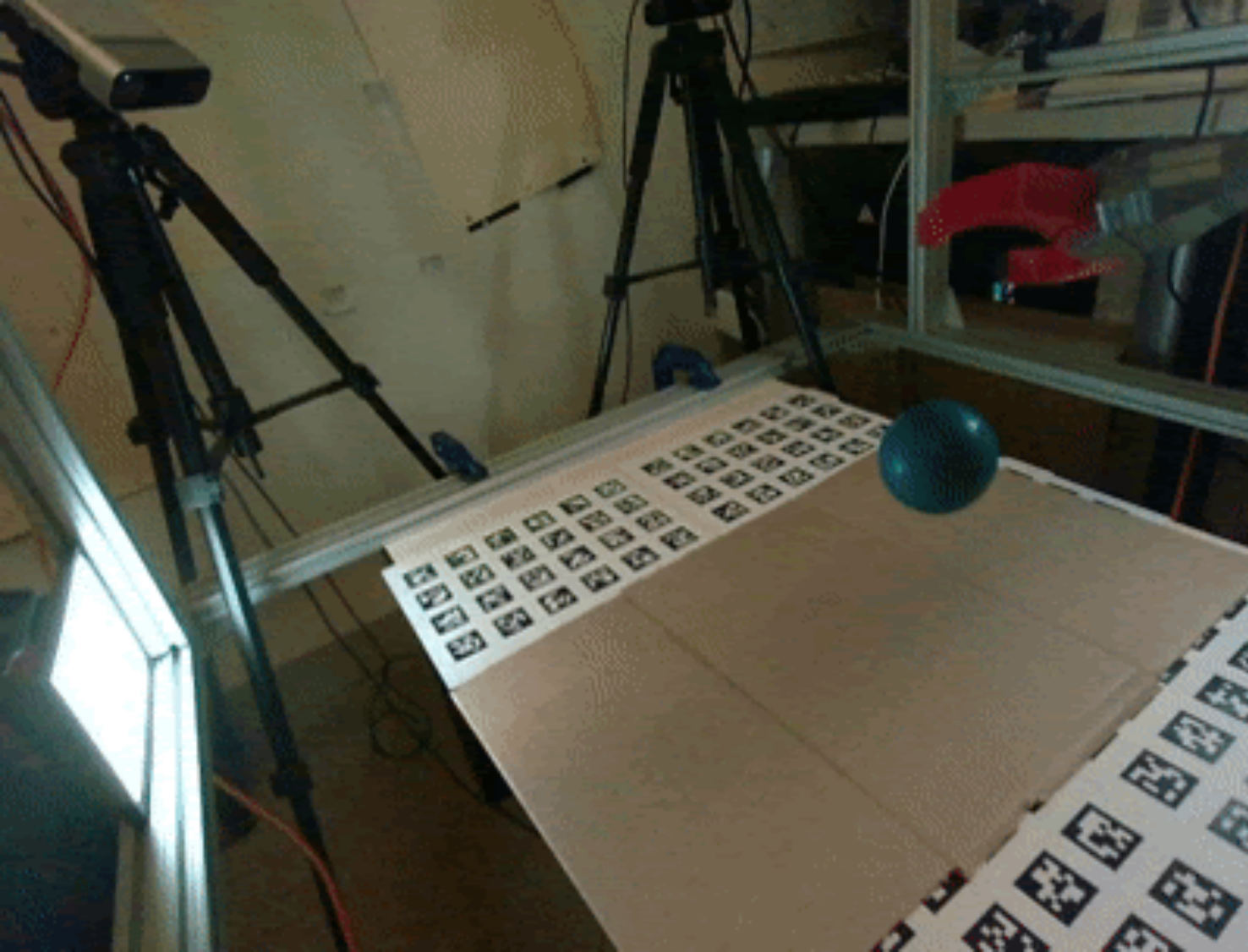


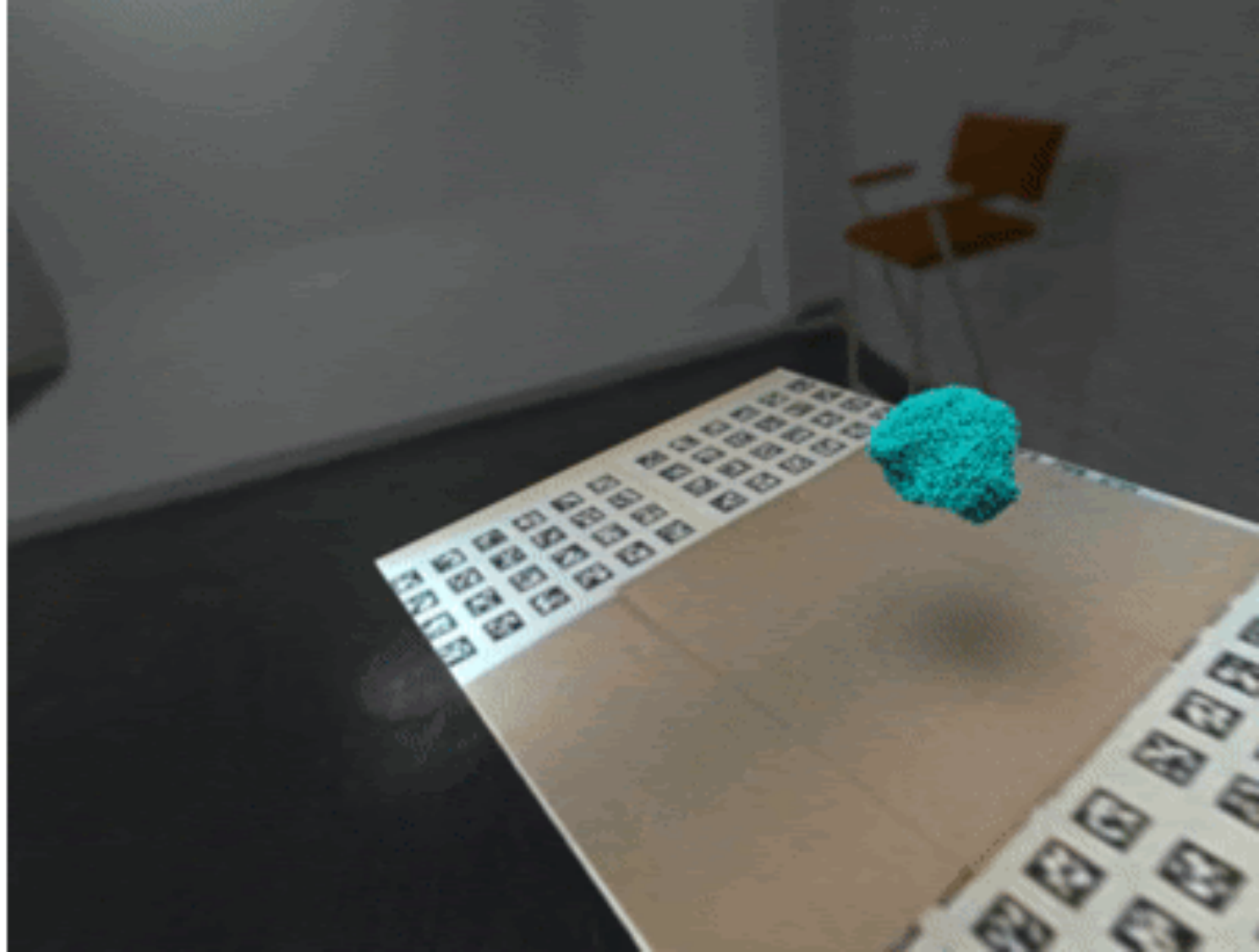
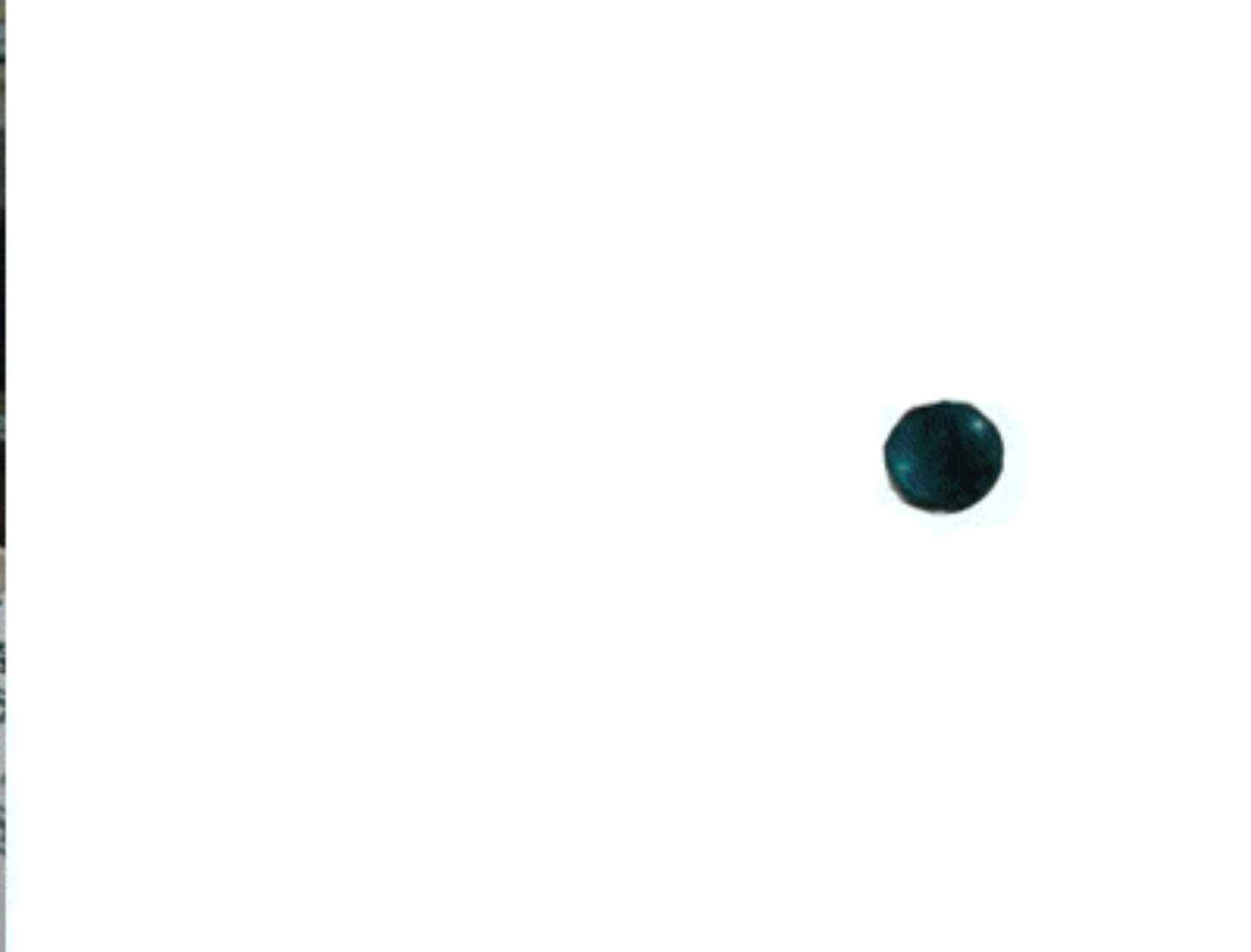
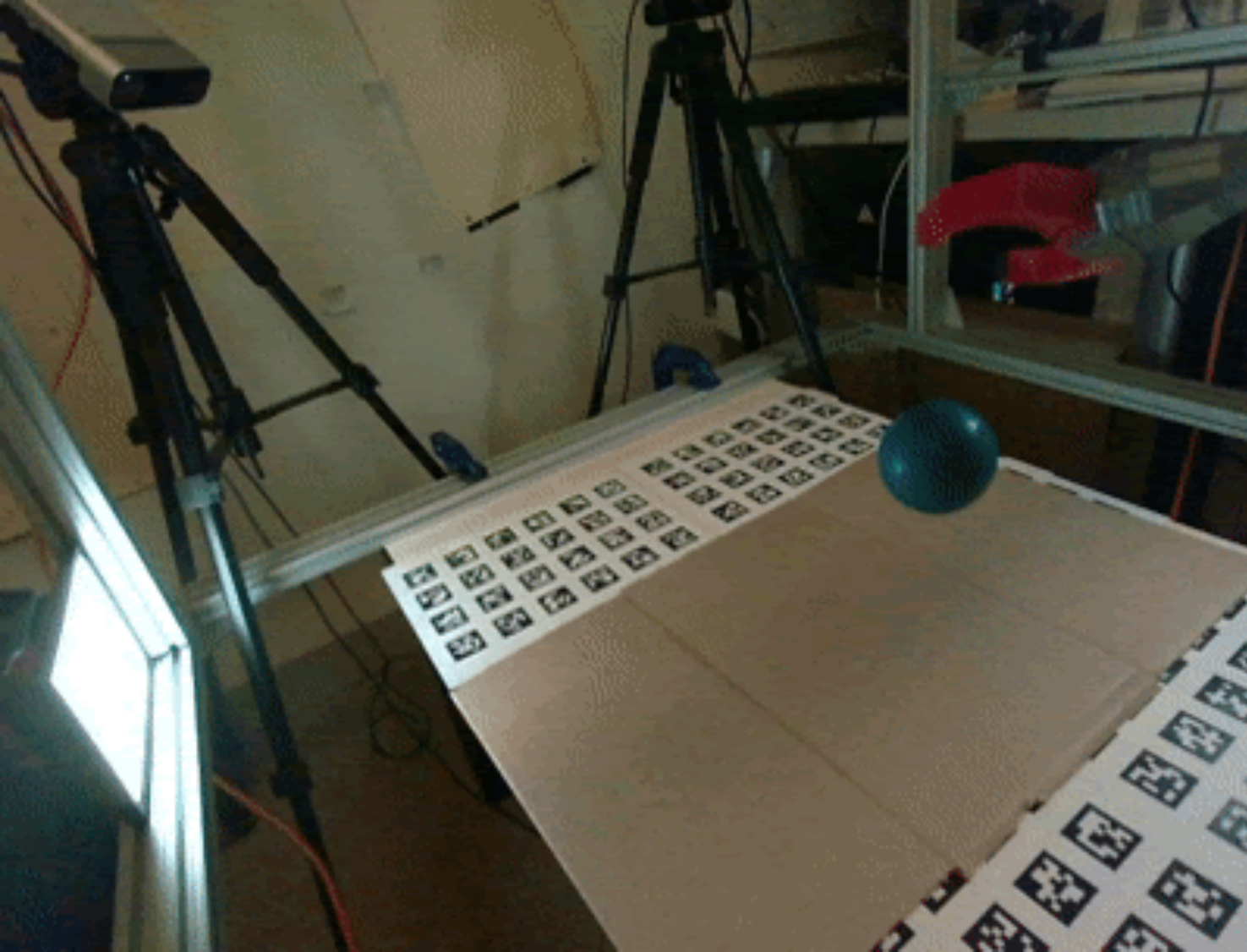


Synthetic Data

Real Data



Ground Truth



PAC-NeRF: Data Collection

Ground Truth

Generated by MLS-MPM framework (Hu et al., 2018b).

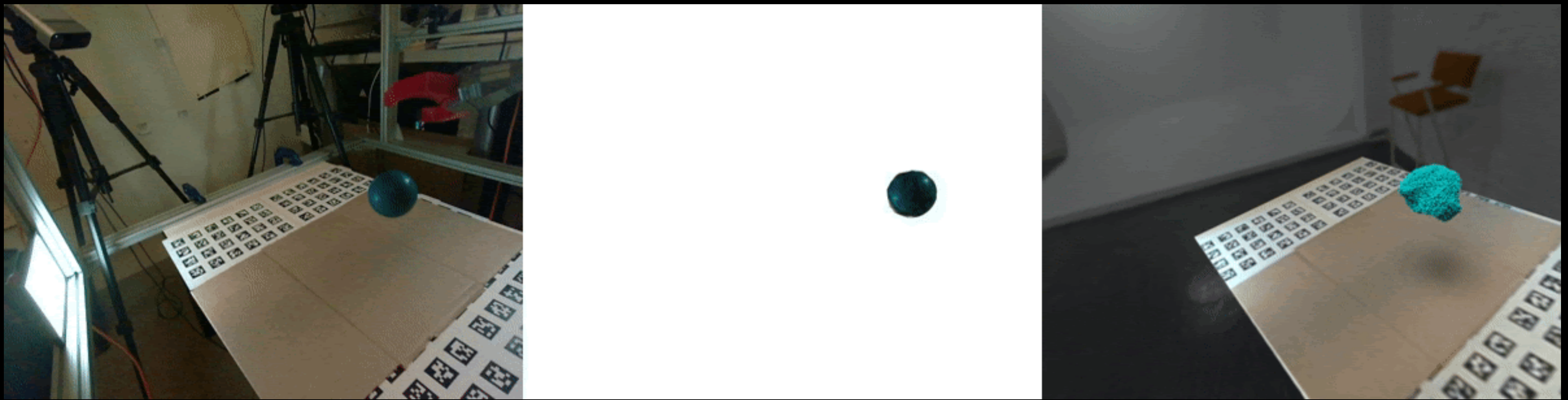
→ Moving least squares material point method
Analyze 9 synthetic problems

Synthetic Data

Real Data

Four synchronized Intel RealSense D455 cameras

→ Very few compared to synthetic data.



PAC-NeRF: Model Architecture

		Ours	Multi-view LSTM	D-NeRF + DiffSim	NeRF + ∇ Sim	Random	Oracle ($\times 10^{-2}$)
Newtonian	$\log_{10}(\mu)$	11.6 ± 6.60	14.1 ± 9.26	> 291.3	Not supported	64.1 ± 31.3	19.4 ± 6.95
	$\log_{10}(\kappa)$	16.7 ± 5.37	28.2 ± 19.8	> 85.2		72.8 ± 42.5	258.9 ± 25.2
	v	0.86 ± 1.45	23.1 ± 17.8	25.5		55.5 ± 22.6	2.82 ± 1.72
Non-Newtonian	$\log_{10}(\mu)$	24.1 ± 21.9	39.4 ± 26.9	> 276.7	Not supported	34.2 ± 19.9	24.0 ± 17.1
	$\log_{10}(\kappa)$	44.0 ± 26.3	25.1 ± 22.6	> 262.5		67.6 ± 49.4	82.90 ± 21.0
	$\log_{10}(\tau_Y)$	5.09 ± 7.41	7.19 ± 7.88	> 359.6		30.0 ± 15.0	9.75 ± 11.3
	$\log_{10}(\eta)$	28.7 ± 23.3	39.9 ± 21.1	> 86.1		64.3 ± 43.7	90.6 ± 37.2
	v	0.29 ± 0.13	24.4 ± 14.2	25.9		52.2 ± 21.6	2.60 ± 0.39
Elasticity	$\log_{10}(E)$	3.02 ± 3.72	17.7 ± 9.25	> 437.5	151.6 ± 42.6	96.2 ± 46.9	4.39 ± 4.54
	ν	4.35 ± 5.08	81.8 ± 58.4	7.67	16.4 ± 6.58	10.9 ± 6.37	3.65 ± 2.88
	v	0.50 ± 0.23	6.10 ± 3.27	30.7	182.4 ± 74.1	43.6 ± 25.4	2.69 ± 0.97
Plasticine	$\log_{10}(E)$	83.8 ± 68.4	41.1 ± 31.4	23.2	Not supported	42.9 ± 38.0	56.2 ± 30.7
	$\log_{10}(\tau_Y)$	11.2 ± 14.5	28.6 ± 17.3	> 268.9		82.7 ± 43.1	8.13 ± 3.61
	ν	18.9 ± 15.7	5.40 ± 3.49	10.36		10.6 ± 6.15	4.44 ± 3.61
	v	0.56 ± 0.17	22.0 ± 14.8	> 384.6		47.7 ± 17.53	4.06 ± 1.87
Sand	θ_{fric}	4.89 ± 1.10	20.1 ± 2.06	64.5	Not supported	22.5 ± 14.7	0.44 ± 0.42
	v	0.21 ± 0.08	14.6 ± 6.25	40.6		38.7 ± 21.1	1.17 ± 0.42